

Label-Efficient Dark-Vessel Detection in SAR: Does SAR-Domain Pretraining Beat Optical and ImageNet Transfer Across ViT and CNN?

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Abstract

Detecting dark vessels—ships without an Automatic Identification System correlate—in spaceborne synthetic aperture radar (SAR) is operationally important, but human-verified examples are scarce. We investigate whether pretraining domain improves label efficiency on xView3-SAR and whether any benefit generalizes across architecture families. Our controlled design uses two size-matched tracks: ViT-B/16 (approximately 86M parameters) and ConvNeXt-V2-Base (approximately 89M). Within each track, we compare random initialization with downloaded encoders trained on generic ImageNet, optical remote-sensing imagery, or SAR imagery. All eight arms share a fixed three-channel [VH, VV, VH–VV] input, point-native heatmap detector, optimizer, schedule, scene-level splits, and seed, and are fine-tuned at nested 10%, 25%, 50%, and 100% training-scene fractions. We measure distance-matched vessel-detection F1 on held-out scenes and, on a disjoint once-touched set of 50 human-verified scenes, dark-vessel recall and near-shore F1. All values are reported as seed-0 point estimates; external YOLO and zero-shot vision-language references remain separate from the controlled curves. By separating initialization comparisons within an architecture from matched-role comparisons across architectures, the study provides a reproducible test of when pretrained representations reduce labeling demand for maritime SAR detection.